



GDCNet: Generative Discrepancy Comparison Network for Multimodal Sarcasm Detection



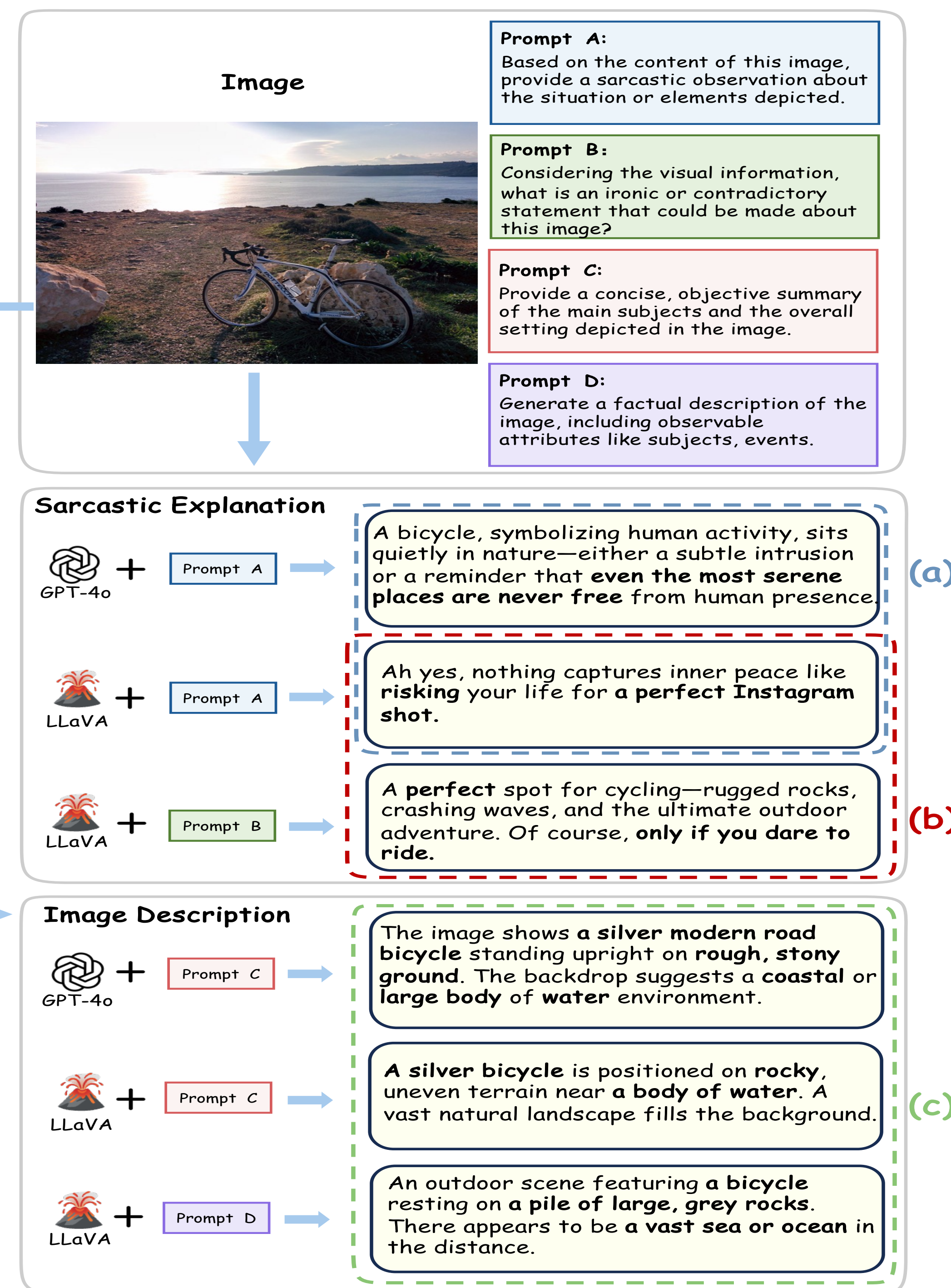
Shuguang Zhang^{1,2}, Junhong Lian^{1,2}, Guoxin Yu^{1,2,3}, Baoxun Xu⁴, Xiang Ao^{1,2,*}

¹ State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences

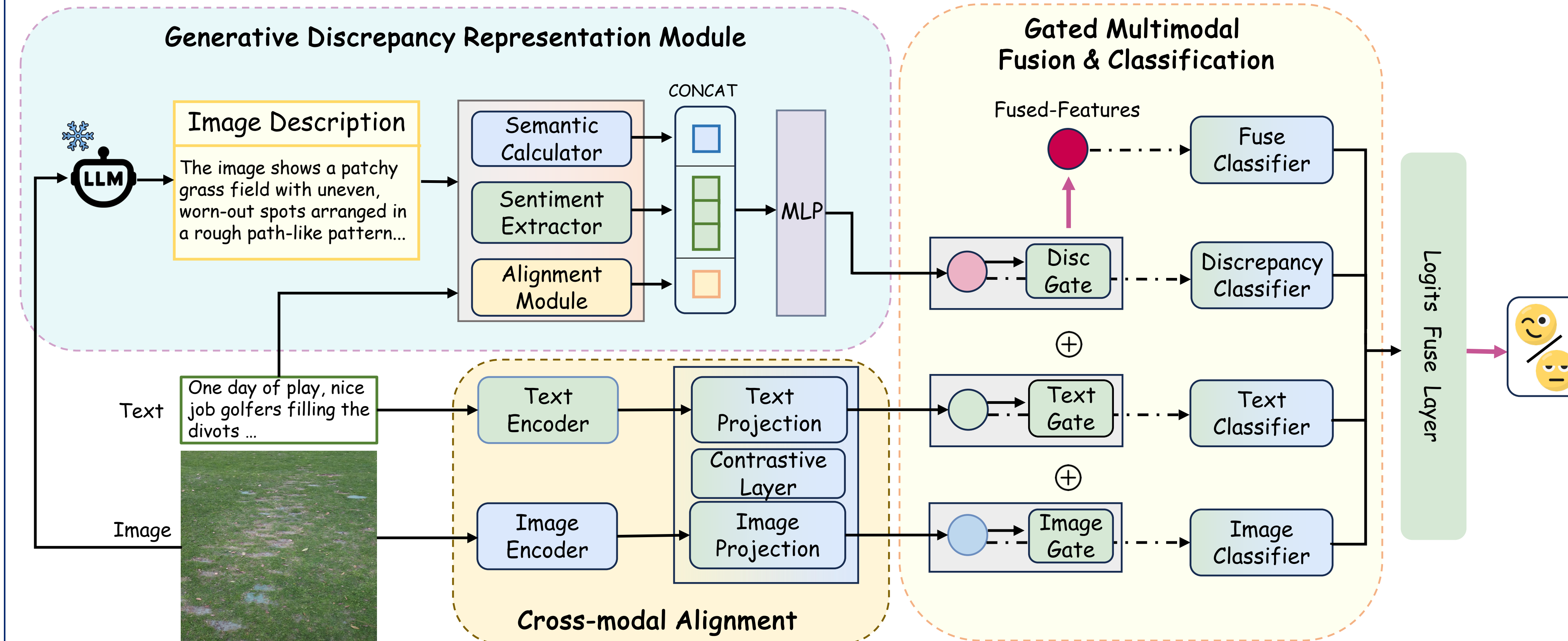
² University of Chinese Academy of Sciences ³ Pengcheng Laboratory ⁴ Shenzhen Stock Exchange

MOTIVATION

- Multimodal sarcasm detection aims to identify sarcasm in image–text pairs.
- Existing methods often rely on coarse cross-modal inconsistency and struggle when image–text relations are weak or implicit.
- Direct sarcastic explanation generation with LLMs/MLLMs is diverse and subjective, introducing noise.
- Key insight:** factually grounded image descriptions are more stable and can serve as semantic anchors.



Methodology: Architecture of GDCNet



- Problem Formulation:** classify an image–text pair (I, T) as sarcastic or non-sarcastic.
- Cross-modal Feature Alignment:** CLIP-based image and text encoders project features into a shared space; contrastive learning aligns matched pairs.
- Generative Discrepancy Representation Module:** an MLLM generates an image-grounded factual description $\hat{T}$; semantic discrepancy, sentiment discrepancy, and visual-textual fidelity are computed.
- Gated Multimodal Fusion & Classification:** gated fusion combines F_T , F_I , and F_D for final prediction.
- Optimization Objective:** jointly optimize binary classification and contrastive alignment.

Ablation Study

Method	Acc. (%)	P (%)	R (%)	F1 (%)
Full Model	87.38	83.39	89.51	86.34
- w/o GDRM	84.42	78.56	86.17	82.19
- w/o SemD	86.23	80.27	87.09	83.54
- w/o SenD	85.98	81.74	87.63	84.58

Removing GDRM causes the largest performance drop, validating the importance of explicit discrepancy modeling.

Main Results on MMSD2.0

Method	Acc.(%)	P(%)	R(%)	F1(%)
Text-Only Methods				
BiLSTM [14]	72.48	68.02	68.08	68.05
SMSD [15]	73.56	68.45	71.55	69.97
BERT [16]	76.52	74.48	73.09	73.78
Image-Only Methods				
ResNet [17]	65.50	61.17	54.39	57.58
ViT [18]	72.02	65.26	74.83	69.72
Multi-Modal Methods				
InCrossMGs [19]	79.83	75.82	78.01	76.90
HKE [5]	76.50	73.48	71.07	72.25
Multi-view CLIP [7]	85.14	80.33	88.24	84.09
DIP [20]	84.63	84.17	85.20	84.68
TFCD [21]	86.54	82.46	87.95	84.31
MoBA [22]	85.01	80.46	87.67	83.64
CofiPara [10]	85.66	85.79	85.43	85.61
ADs [23]	85.60	85.28	85.60	85.41
GDCNet (Ours)	87.38	83.39	89.51	86.34

Comparison with LLM-Based Methods

Method	Acc. (%)	P (%)	R (%)	F1 (%)
LLaVA (Zero-Shot)	51.06	40.09	46.40	43.02
LLaVA (CoT)	48.69	40.93	65.17	50.28
Qwen-VL (Zero-Shot)	40.63	32.44	35.53	33.63
Qwen-VL (CoT)	58.86	56.82	58.67	57.26
GPT-4o (Zero-Shot)	71.07	79.52	71.07	70.24
GPT-4o (CoT)	74.26	65.81	72.68	68.92
GDCNet(Ours)	87.38	83.39	89.51	86.34

GDCNet substantially outperforms direct zero-shot and CoT prompting of general-purpose MLLMs.

Conclusion

GDCNet mitigates noise from subjective sarcasm generation by using image-grounded factual descriptions as cross-modal anchors. By explicitly modeling semantic discrepancy, sentiment discrepancy, and visual-textual fidelity, and integrating them with gated multimodal fusion, GDCNet achieves robust multimodal sarcasm detection and establishes a new state of the art on MMSD2.0.